

वन VANA

said like “one” · Hindi for “forest”

Your AI Mental Health Companion

VANA is a calm, voice-first companion for people carrying stress, anxiety or sadness. You speak or type in English, Hindi, Marathi or Hinglish. VANA answers gently in your language, with soft forest sounds in the background, and saves the conversation privately so you can come back to it. When a message signals real danger, VANA stops and shows verified Indian helplines instead of improvising.

Frontend React + Vite

API Go on Vercel

AI Groq · Google Gemini

Data Supabase Postgres

Reading time about 15 min

Stay safe →

How VANA handles a crisis

See it work →

Asha's first conversation, step by step

Ship it →

The exact steps to put version-6 on stage

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- 04 Screen-by-screen tour
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In crisis? Tele-MANAS **14416** or 1-800-891-4416 (free, 24×7). Emergency **112**.

So many people carry it *alone*

Most people who feel low never talk to anyone about it. The reasons are ordinary, and they stack up.

SILENCE

It feels too small to mention

Stress, worry and sadness get carried quietly, day after day, until they are heavy.

REACH

Care feels far away

Therapy can be costly, booked out, or simply hard to ask for in your own words.

2:00 AM

No one is awake

The hardest moments tend to arrive late at night, when friends and clinics are asleep.

LANGUAGE

Feelings live in your own language

Many people think in Hindi or Marathi, or a mix with English, and most apps answer only in English.

WHAT VANA IS FOR

A safe first place to put feelings into words, at any hour, in your language. It helps you calm down and think clearly, and it points you to real people when you need more than an app.

A calm companion that *listens*

FEATURE	WHAT IT MEANS FOR YOU
Voice or text	Press <i>Speak to VANA</i> and talk, or type. Voice input is cleaned up (filler words, mis-heard words) before VANA reads it.
Your language	Write in English, Hindi, Marathi or Hinglish. VANA replies in the same language and the same script: Hinglish typed in English letters gets English letters back.
Forest soundscapes	Choose <i>Deep Forest</i> , <i>Rain Sanctuary</i> or <i>Mossy Stream</i> when you arrive. The sound dips while you speak and changes gently with the mood of the conversation.
Gentle, practical replies	A short heading, a few simple steps you can try now, and a one-line nature image at the start and end. No drug names or dosages, ever.
Private history	Every conversation is saved to your account. Open any of them from <i>Recent Sessions</i> and carry on.
A safety net	Messages that signal danger skip the AI completely and get a fixed reply with verified helplines. See The safety net .
Three themes	Greening (the default forest look), Dark and Light.

Asha's first evening with VANA

Asha works long hours and can't switch off at night. This is everything that happens, on screen and behind it, from sign-up to a saved conversation.

1

She creates an account

Name, email and a password of at least 8 characters. The server stores only a **bcrypt hash** of the password, never the password itself, and signs her in with a token that lasts 7 days.

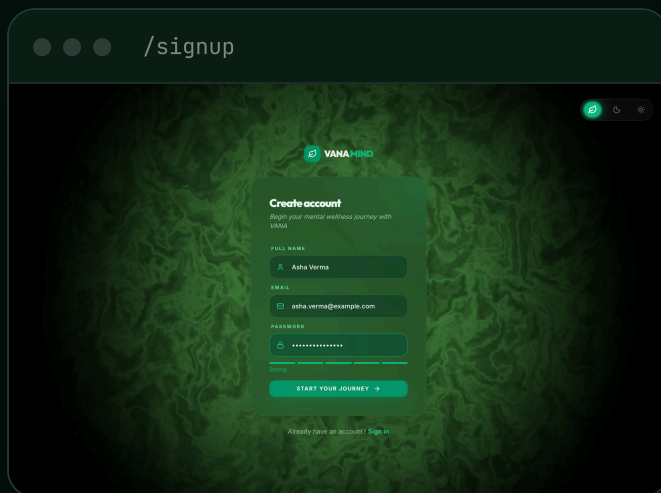


Fig. 1 Sign-up, in the default Greening theme.

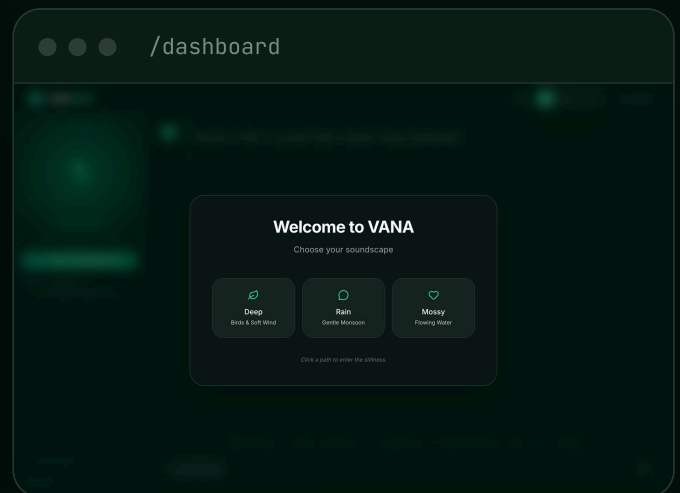


Fig. 2 The first thing she sees: choose a soundscape.

2

She chooses a soundscape

Deep Forest: birdsong and soft wind. Choosing also unlocks audio in the browser, which is why VANA asks first instead of playing sound on its own.

3

She writes what's on her mind

"I've been feeling really stressed at work, and I can't switch off at night."

Behind the scenes, the server checks the message against the crisis keywords first. Nothing matches, so it creates a new conversation for her, titles it with her first words, and asks the fast model on Groq for a reply. Groq usually answers in 2 to 5 seconds. If it's busy, Google Gemini answers instead.

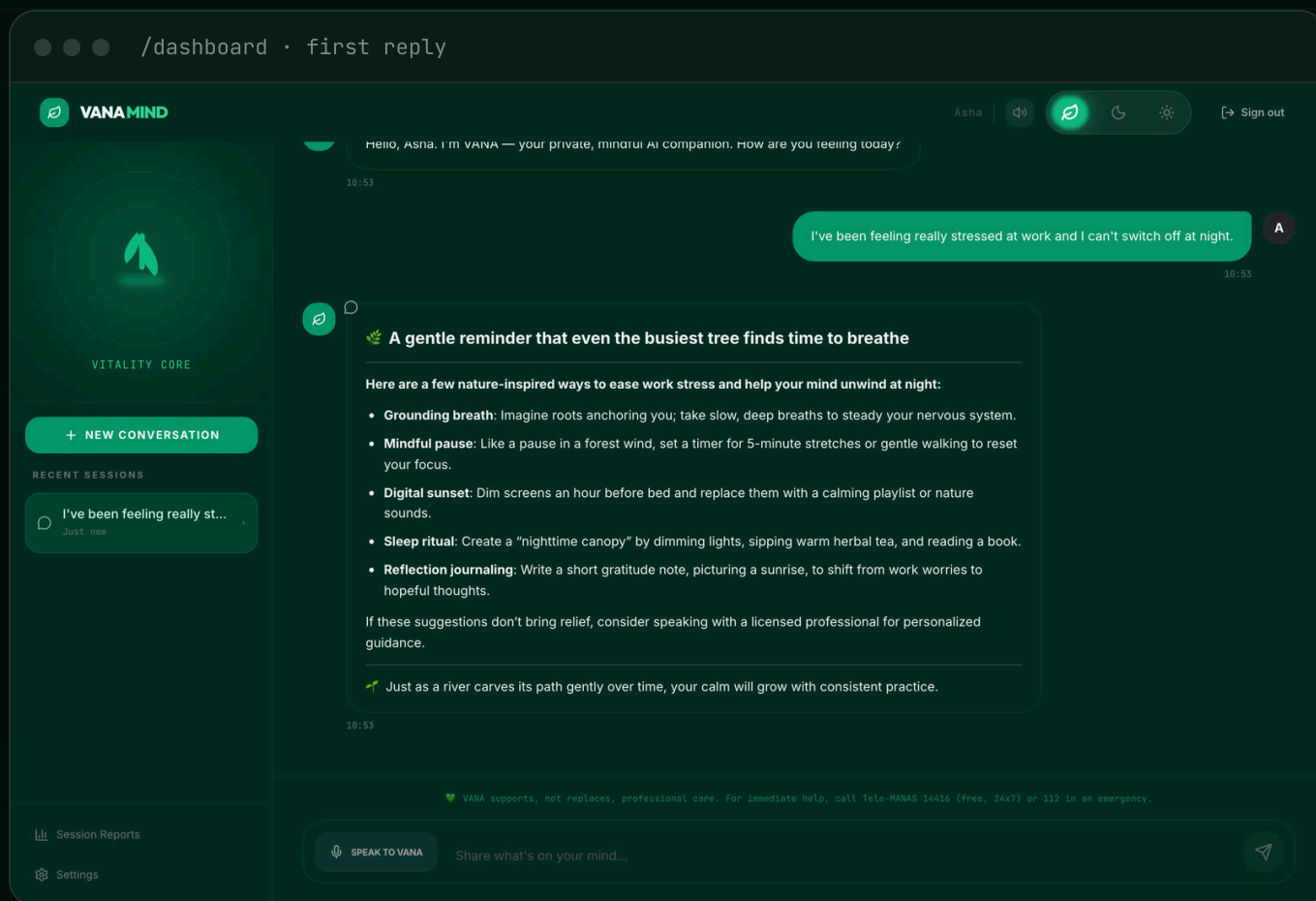


Fig. 3 The reply: a heading, five simple steps, and a nature image at the start and end. The conversation appears in *Recent Sessions* straight away.

4

Both messages are saved together

Her message and VANA's reply are written to the database in one transaction, with the model used, response time and sentiment. If she closes the tab, the conversation is still there tomorrow.

5

She comes back to it

She asks a follow-up. VANA reads the last few messages of *this* conversation for context, so the answer builds on what she already said.

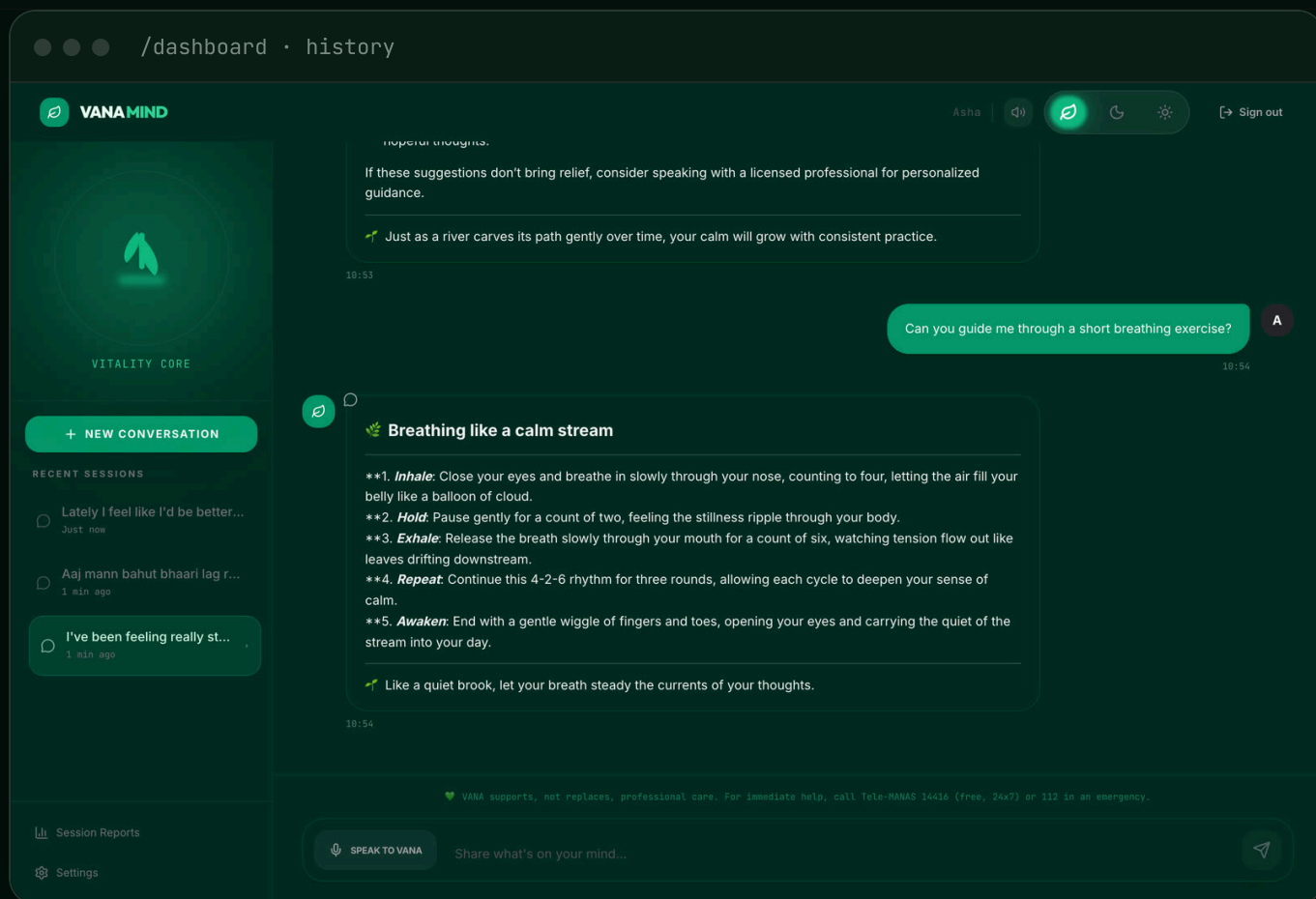


Fig. 4 A reopened conversation, loaded from the database. Three saved sessions in the sidebar.

THE RESULT

In under a minute Asha has calm, practical help in her own words, and a private record she can return to. She never had to explain herself twice.

Every screen, and what it's for

SCREEN	USE IT TO
Sign in <code>/login</code>	Sign in with email and password. A wrong password shows “Invalid email or password.” and nothing else, so nobody can learn which emails have accounts.
Create account <code>/signup</code>	Register with name, email and a password of 8 to 72 characters.
Dashboard <code>/dashboard</code>	Talk to VANA, start a <i>New Conversation</i> , reopen one from <i>Recent Sessions</i> , switch theme, mute sound, sign out.

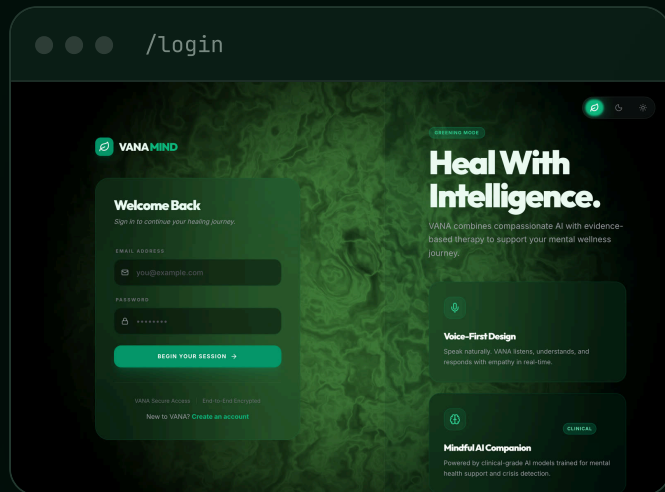


Fig. 5 Sign in.

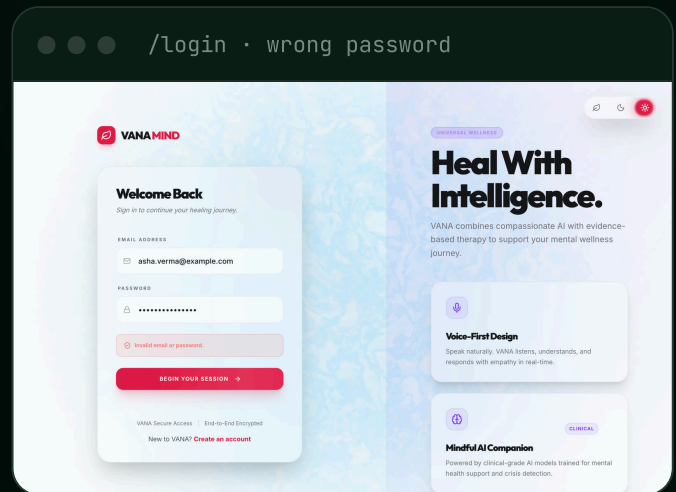


Fig. 6 A wrong password is refused with one clear message.

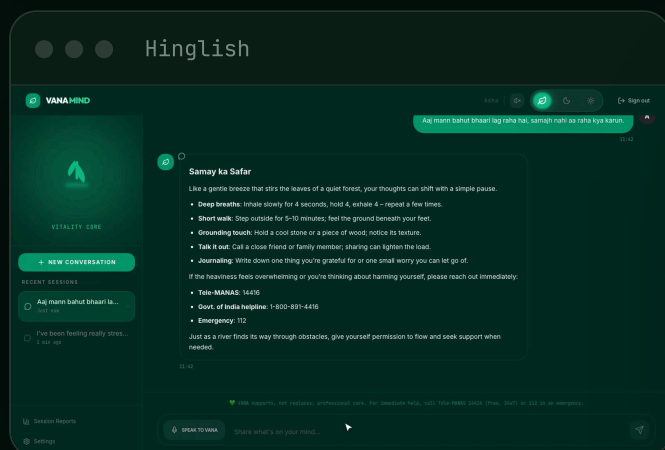


Fig. 7 Hinglish in English letters gets English letters back.

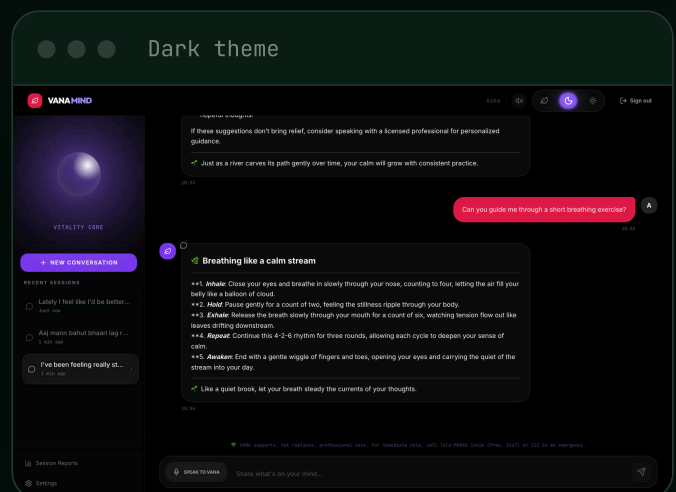


Fig. 8 The Dark theme.

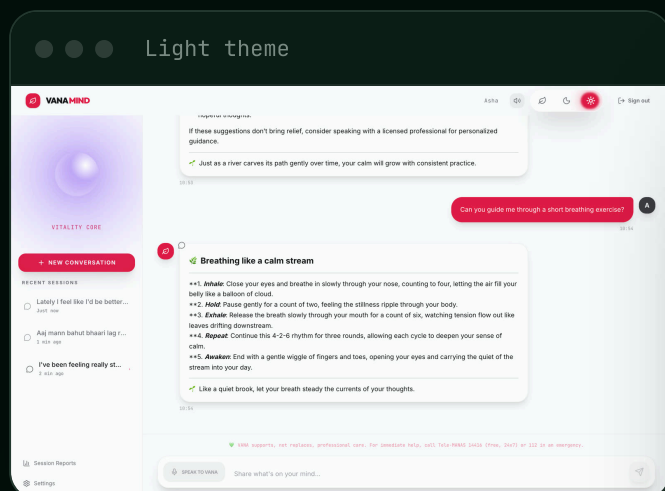


Fig. 9 The Light theme.

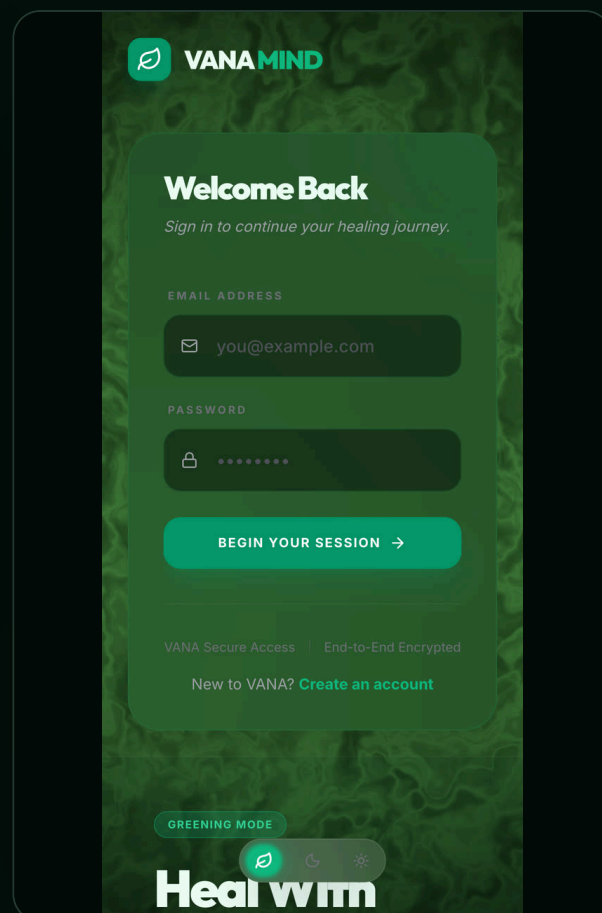


Fig. 10 On a phone.

When it matters most, VANA *doesn't improvise*

Every message is checked against a list of crisis phrases before any AI sees it. A serious match skips the AI completely and returns a fixed, human-written reply with exact helpline numbers. The conversation is flagged for review.

14416

Tele-MANAS, Govt. of India.

Free, confidential, 24x7, in your language.

1-800-891-4416

Tele-MANAS toll-free number.

112

Emergency services, or go to the nearest hospital emergency.

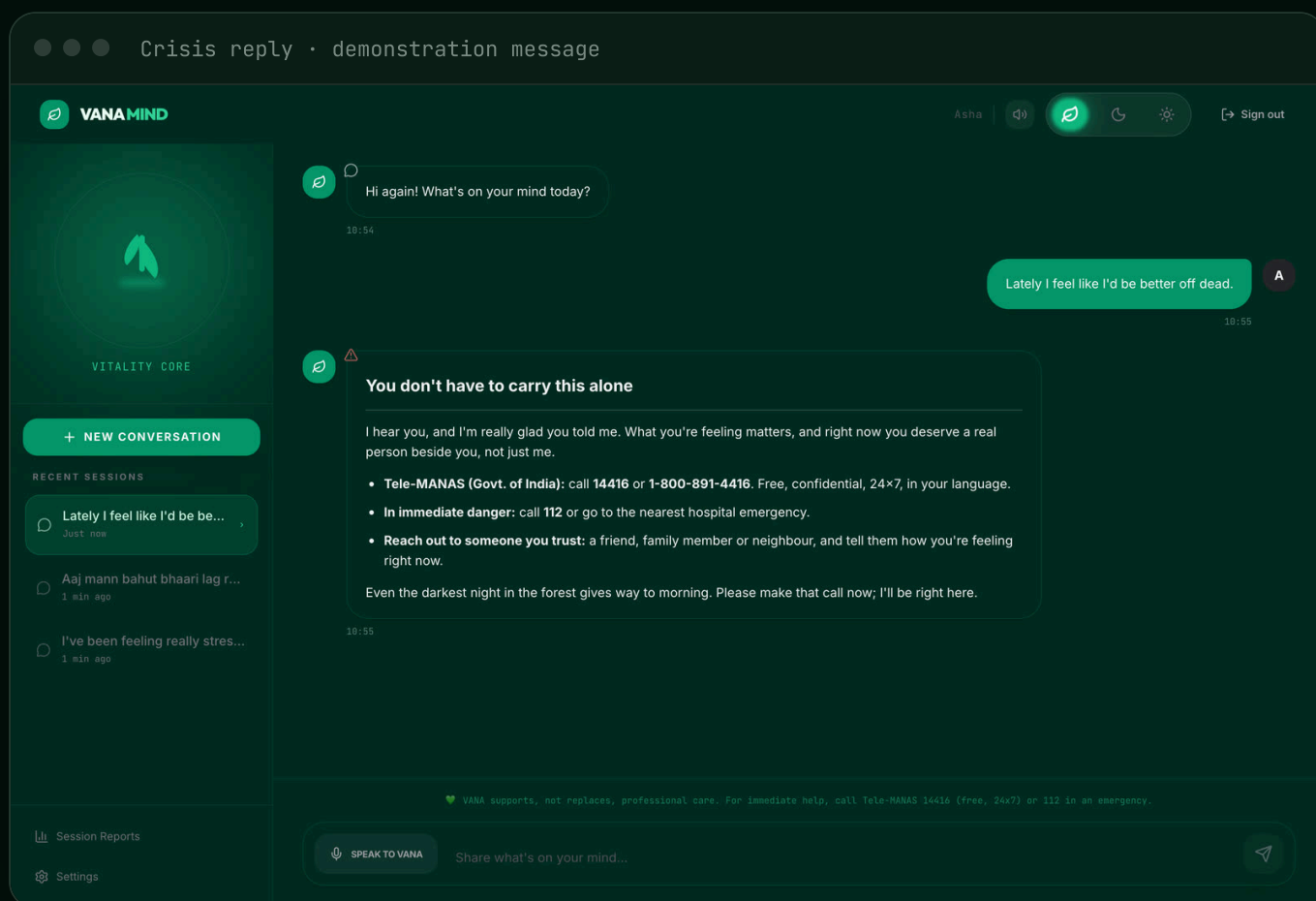


Fig. 11 The crisis reply, triggered here by a demonstration message. The wording and numbers are fixed in code, so they are always exact.

How a message is judged

KEYWORD SEVERITY	EXAMPLES	WHAT VANA DOES
Critical / High	suicide, kill myself, end my life, want to die, better off dead	Fixed crisis reply with helplines. Conversation marked crisis_detected and the message flagged_for_review .
Medium	hopeless, worthless	A normal reply, but the AI is told to validate first, answer gently and mention Tele-MANAS.
Low	help	A normal reply. "Can you help me sleep?" is not a crisis.

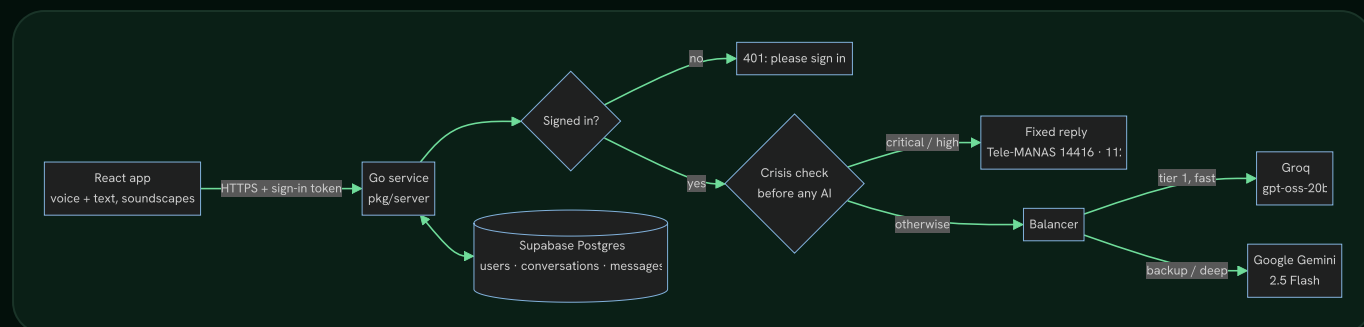
Phrases match as whole words with common endings, so “suicidal”, “suicides” and “overdosed” are caught, while “helpful” and “want to diet” are not. Asking for songs, films or a doctor near you counts as a coping request, not a crisis.

ONE MORE GUARD

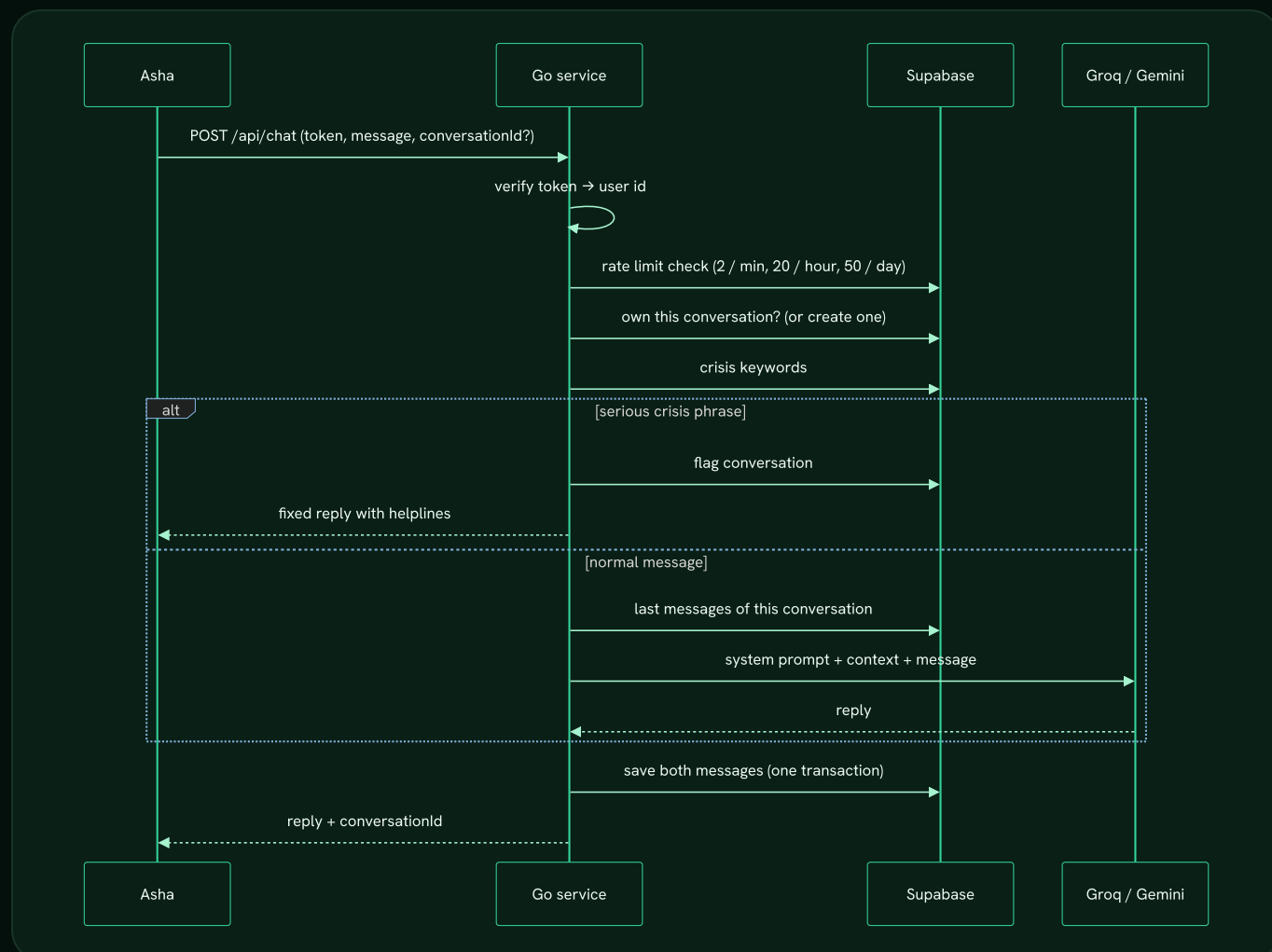
The AI is only allowed to write three phone numbers: 14416, 1-800-891-4416 and 112. In testing before this rule, the model once invented a “national helpline” number. That can't happen silently any more.

Simple, by *design*

A React app in the browser talks to one small Go service. On Vercel that service runs as a serverless function; on a laptop it runs as a normal server. Both use exactly the same code.



What happens during one message



The API

ROUTE	SIGN-IN	DOES
POST /api/auth/register	No	Create an account. Returns a token and the user.
POST /api/auth/login	No	Check the password. Returns a token and the user.
GET /api/auth/me	Yes	Confirm a stored token is still valid.
POST /api/chat	Yes	Send a message. Creates or continues a conversation and saves both messages.
GET /api/conversations	Yes	Your 50 most recent conversations, newest first.
GET /api/conversations/{id}/messages	Yes	Every message in one of your conversations.
GET /api/health	No	Liveness check.

Errors always come back as `{"success": false, "message": "..."}` , and the app shows that message to the user as written.

Where things are in the repository

PATH	CONTAINS
pkg/server/	Every HTTP route, sign-in, chat persistence. Shared by local and Vercel.
pkg/orchestration/	Crisis check, prompts, Groq and Gemini clients, the balancer.
pkg/database/	All SQL: users, conversations, messages, keywords, rate limits.
api/index.go	The Vercel entry point (a few lines).
main.go	The local server entry point (a few lines).
src/	The React app: <code>pages/</code> , <code>hooks/useAuth.tsx</code> , themes.
docs/migrations/	<code>006_password_auth_and_history.sql</code> for version-6.

Before and after

A full check of the live app found these problems. Each one is fixed on the `version-6` branch and covered by tests.

Signing in

BEFORE

Any password, or no password at all, signed you in. The email alone was enough.

NOW

Passwords are stored as bcrypt hashes and checked on every sign-in. Wrong or missing passwords get “Invalid email or password.”

BEFORE

The chat endpoint trusted a `userId` sent in the request, with no token.

NOW

Your identity comes only from a verified, signed token. Another person's conversation returns “not found”.

Saving conversations

BEFORE

No message was ever saved. The sidebar showed four made-up sessions, and rate limits never applied.

NOW

Every exchange is saved, the sidebar lists your real conversations, and any of them can be reopened.

Safety

BEFORE	Any message containing “help” triggered the crisis reply, and the helplines shown were US numbers.
NOW	Only serious phrases trigger it, the numbers are Indian (14416, 112), and the AI can't make up numbers.
BEFORE	Hinglish typed in English letters often got a reply in Devanagari.
NOW	Replies match your script. A mismatched reply is regenerated once automatically.

Tests that pass

CHECK	RESULT
Go unit tests: token rules, validation, crisis matching, script detection	all pass
End-to-end API run against the real database (register, login, chat, history, isolation, crisis)	35 / 35
Browser run of the real UI (sign up, chat, reload, reopen, sign out, wrong password, sign in)	9 / 9
Hinglish replies in English letters	8 / 8

Four tables do the work

TABLE	HOLDS
users	Name, email (unique, any case), <code>password_hash</code> , last sign-in.
conversations	Owner, title (the first words), message count, crisis flag and reason.
messages	Every message: who sent it, text, sentiment, model, response time, crisis and review flags.
crisis_keywords	17 phrases with a severity of low, medium, high or critical.
clinical_anchors	Reference prompts, including the crisis protocol with Indian helplines.
api_usage_logs	One row per chat request, for monitoring.

Deleting a user deletes their conversations and messages with them. Row-level security is on for every table, so the public Supabase key can't read anything; only the Go service can.

Two terminals, one browser

You need Go 1.22 or newer, Node 20 or newer, and a `.env` file in the project root (copy `.env.example`).

`.env` (required values)

```
DATABASE_URL=postgresql://...pooler.supabase.com:6543/postgres
JWT_SECRET=# 32+ random characters: openssl rand -base64 48
GROQ_API_KEY=gsk_...
GROQ_MODEL=openai/gpt-oss-20b
GEMINI_API_KEY=AIza...
GEMINI_MODEL=gemini-2.5-flash
```

terminal 1 · API on :3000

```
git checkout version-6
go run .
```

terminal 2 · app on :5173 (proxies /api to :3000)

```
npm install
npm run dev
```

Open `http://localhost:5173`, create an account and start a conversation. Run the tests with `go test ./...`.

THE SERVER REFUSES WEAK SECRETS

If `JWT_SECRET` is missing, shorter than 32 characters or a placeholder such as `your-super-secret...`, sign-in is switched off and the log says so. A guessable secret would let anyone forge a sign-in.

Five steps to put version-6 live

1

Database: already done

Migration `006_password_auth_and_history.sql` has been applied to the Supabase database. It only adds things (a column, indexes, keywords), so the current live version keeps working. It is safe to run again.

2

Set a strong `JWT_SECRET` in Vercel

Vercel → VANA → Settings → Environment Variables. Set it for **Preview** and **Production**. Generate one with `openssl rand -base64 48`. If the current value is a placeholder, sign-in will stay off until this is set.

3

Push the branch

```
git push -u origin version-6
```

Vercel builds a preview from the branch. Open a pull request into `main`.

4

Check the preview

Create a new account, chat, reload the page and reopen the conversation, sign out, try a wrong password, then sign in. In Vercel → Logs there should be no `FATAL CONFIG` line.

5

Merge and redeploy

Merge the pull request. Vercel redeploys production from `main`.

EXISTING ACCOUNTS NEED A NEW PASSWORD

The 5 accounts created before version-6 have no password stored, so they can't sign in. They are test accounts from May. Either delete them and sign up again, or set a password for them directly.

When something goes wrong

“Invalid email or password.” for an old account

Accounts made before version-6 have no password stored. Delete the account and sign up again, or set a password hash for it directly.

Sign-in says “Authentication is not configured on the server.”

`JWT_SECRET` is missing, too short or a placeholder. Set a random value of 32+ characters and restart or redeploy.

“You're sending messages quickly...”

VANA allows 2 messages a minute, 20 an hour and 50 a day per person. Wait a minute and try again.

“VANA is waking up. Please try again in a moment.”

The database didn't answer. Check that the Supabase project is running (free projects pause after about a week without activity) and that `DATABASE_URL` is right. The service reconnects on its own once it's back.

Replies are slower than usual

The Groq key is on the free tier (8,000 tokens a minute). When it's full, VANA uses Gemini instead, which is slower but still answers. Upgrading the Groq tier removes this.

No sound

Choose a soundscape when the dashboard opens; browsers block audio until you click. Check the speaker button in the top bar isn't muted.

What's next

- **Password reset.** There is no “forgot password” email yet.
- **Clinical knowledge.** The `clinical_knowledge` table is empty, so answers don't draw on reviewed reference material yet. Searching is wired up and ready.
- **Groq capacity.** The free tier fills up quickly with more than a few people chatting.
- **Supabase pausing.** Free projects pause after inactivity. Upgrade, or schedule a daily sign-in check against stage.
- **Session reports and settings.** The sidebar buttons are placeholders.
- **Professional review.** The crisis wording and helplines should be reviewed by a mental-health professional before a public launch.

Words used in this guide

VANA (वन)	Hindi for forest. Said like the English word “one”.
TeLe-MANAS	India's free national tele-mental-health service. 14416 or 1-800-891-4416, 24x7.
Hinglish	Hindi and English mixed, usually typed in English letters.
bcrypt hash	A one-way fingerprint of a password. It can check a password but can't be turned back into one.
Token (JWT)	A signed pass the app sends with each request to prove who you are. Lasts 7 days.
Groq	A fast AI service. VANA's first choice for replies.
Gemini	Google's AI model. VANA's backup and deeper-answer model.
Supabase	Hosted Postgres database where accounts and conversations live.
Soundscape	The background nature sound: Deep Forest, Rain Sanctuary or Mossy Stream.